

Measles

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Measles (Rubeola)

Prof Onyire Nnamdi Benson – Paediatrics, topic 37 · Clinical Features · Complications · Control

Outline

Epidemiology & local context · Pathogenesis & clinical course · Clinical features · Complications · Diagnosis · Management · Prevention · Key takeaways

Epidemiology & local context

The agent is the **measles virus** – a **Morbillivirus** of the **Paramyxoviridae** family; an **RNA virus with one serotype**. Transmission is by **airborne droplets** and it is **extremely contagious** – about **90% of non-immune contacts** become infected. Measles remains **endemic in Nigeria** despite the vaccine, with **2022-2023 outbreaks in Kano, Sokoto, Borno and Zamfara** driven by low coverage and insecurity. **High-risk groups** are children under 5, the malnourished, HIV-positive and vitamin-A-deficient; the **case fatality is 3-6% in Nigeria versus < 0.1% in the US**. Seasonality is the **dry season (Nov-April)** in the Sahel, including Northern Nigeria.

Nigeria fact: measles is one of the **6 childhood killer diseases** in Nigeria’s NPI + routine-immunisation schedule.

Pathogenesis & clinical course

Pathogenesis: respiratory mucosa -> lymphatics -> **viraemia** -> epithelial infection -> **rash + immunosuppression for 4-6 weeks**.

Clinical stages:

- **Incubation** – 10-14 days (average 12).
- **Prodrome / catarrhal** – 2-4 days of **high fever + the “3 Cs”**: **Cough, Coryza, Conjunctivitis**. **Koplik spots** (pathognomonic) appear on **day 2-3**: white/blue spots on the buccal mucosa opposite the molars.
- **Exanthem** – day 3-4. The **maculopapular rash starts on the face -> neck -> trunk -> limbs**, is confluent, and lasts 5-6 days; **fever spikes with the rash**.

Infectious period: the patient is infectious from **4 days before to 4 days after** rash onset.

Clinical features

The classic picture is **fever + the 3 Cs + Koplik spots + a descending maculopapular rash**. Koplik spots – white/blue spots on the buccal mucosa opposite the molars – are **pathognomonic** and precede the rash.

Complications – the real killers

Common: otitis media (7-9%), **pneumonia (1-6%)** – the **#1 cause of death**, diarrhoea, and worsening malnutrition.

Rare but fatal: **encephalitis** (1/1000 -> 15% mortality), **SSPE** (subacute sclerosing panencephalitis, 1/10,000 -> fatal 6-8 years later), **keratitis -> blindness** (especially with vitamin A deficiency), and reactivation of secondary **TB**.

Nigeria-specific: measles + malnutrition + vitamin A deficiency = high case fatality. Measles destroys the gut and immune memory -> **“immune amnesia”**, raising all-cause mortality for 2-3 years afterwards.

How do we diagnose?

Clinical diagnosis: fever + 3 Cs + Koplik spots + descending rash = classic (but atypical in the vaccinated / partially immune).

Lab confirmation – WHO standard:

- **Serology** – measles **IgM** in serum. The gold standard; collect within **4-28 days** of rash.
- **PCR** – measles RNA from throat/nasopharyngeal swab or urine; best in the **first 3 days**.
- **Viral culture** – not routine.

Differential diagnosis: rubella, roseola, dengue, drug reaction, Kawasaki disease. The key discriminator is **Koplik spots + the 3 Cs**.

Management – no antiviral, so we support

There is **no specific antiviral**, so care is supportive:

- **Isolate** – airborne precautions; admit if severe/complicated.
- **Hydration + nutrition** – ORS, continue breastfeeding.
- **Treat complications** – antibiotics for pneumonia/otitis, antipyretics.

- **Vitamin A** (WHO/Nigeria policy) – **200,000 IU orally x 2 doses 24 hours apart** for all children ≥ 8 months (100,000 IU if 6-11 months). This **reduces mortality by ~50%** in deficient areas like Nigeria.
- **Post-exposure prophylaxis** – vaccine within **72 hours**, or immunoglobulin 0.25 ml/kg within **6 days** for high-risk contacts.

Prevention – vaccines & Nigeria’s strategy

Prevention = **vaccination** with the **live attenuated measles vaccine**, given at **9 months and 15 months** in Nigeria’s routine schedule (as the MR vaccine). **Efficacy:** 85% at 9 months, 95% at 12 months; two doses give **97% protection**. **Herd immunity** requires **95% coverage** to stop transmission.

Nigeria’s approach: routine immunisation (9m + 15m MR, free at PHCs); **supplementary immunisation activities (SIAs)** every 2-4 years for children 9-59 months (last national SIA 2022); and outbreak response with case-based vaccination within a 72-hour radius.

WHO target: measles elimination by 2030; Nigeria is still in the “**control phase**.”

Key takeaways for final-year exams / clinic

- **Classic triad:** fever + 3 Cs + Koplik spots = measles until proven otherwise.
- **Deadly Nigerian combo:** measles + malnutrition + vitamin A deficiency -> **always give vitamin A**.
- **Complication to fear: pneumonia** – the #1 cause of measles death.
- **Public health:** 1 case = an outbreak; notify the LGA and isolate immediately – it is **airborne**, not just droplet.
- **“Immune amnesia”:** measles wipes immune memory for 2-3 years, raising all-cause mortality afterwards.
- **OSCE tip:** ask for vaccination history + vitamin A supplementation; check the buccal mucosa for Koplik spots.

Bottom line: “Measles kills through complications, not the rash.”

References

Lecture material of Prof Onyire Nnamdi Benson (500-level Paediatrics, Abia State University); WHO measles guidelines; Nelson Textbook of Paediatrics.